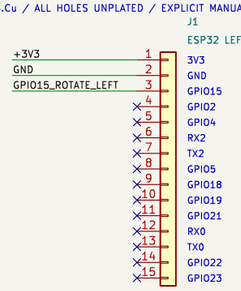
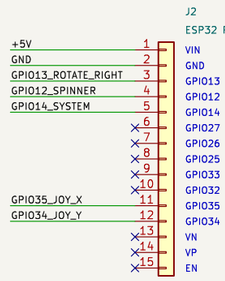
METAL MONSTER / ENCLOSURE SHOULDER BUTTONS / PROVISIONAL CONTROLLER PCB

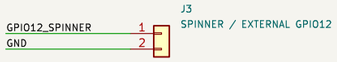
USB TOP / ANTENNA BOTTOM / SOCKET PADS 1..15 TOP TO BOTTOM / 27 mm ROW CENTRES

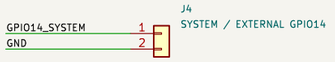
REMOVABLE ESP32 / 3.3 V ANALOG / DRY CONTACT SWITCHES

AVENGERS: F.Cu + B.Cu / ALL HOLES UNPLATED / EXPLICIT MANUAL LAYER LINKS / TOP LEGEND



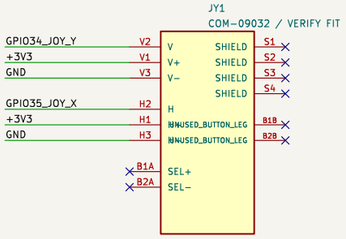






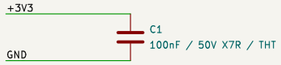






JY1 H2 = X / GPIO35; V2 = Y / GPIO34. H1/V1 = 3V3; H3/V3 = GND.

B1A/B2A/B1B/B2B: all joystick push contacts NC. Spinner uses J3, not JY1.



GPIO12 HIGH at reset can select 1.8V flash supply. J3 button to GND; no external pull-up. GPIO15 LOW mutes boot log.

Check X/Y outer-terminal polarity against forward/backward behavior in trans.ino.

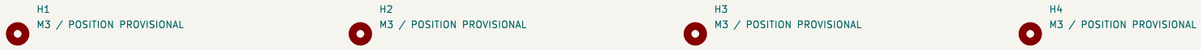
External 2S BMS must monitor BOTH cells, protect and balance; separate 8.4V 2S charger.

No TP4056 pack charger. Cell voltage/current limits, enclosure and antenna clearance remain unverified.

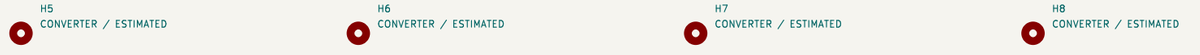








H1..H4 carrier holes provisional; H5..H8 converter holes are USER ESTIMATES, not measured.

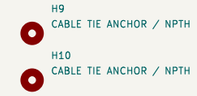


POWER / EXTERNAL PROTECTION AND CHARGER



Protected 2S P+ -> external fuse -> power switch -> J7 BAT+ hole -> converter IN+.

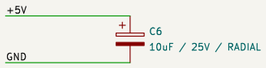
Protected P- -> J7 BAT- hole -> IN-. J7 is mechanical only; sleeve and strain-relieve wires.





J8.1 = OUT+ / 5V; J8.2 = OUT- / load GND. No carrier BAT_NEG; never bridge IN-/OUT-.

Module 51 x 26.3 x 14; H5..H8 estimate 47 x 23 (horizontal) / D3 / 5 mm insulating spacers.

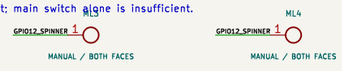


XL4015 IC specified VIN >= 8V; 2S nominal 7.4V. Replacement decision REQUIRED.

Candidate: Pololu S13V30F5 5V buck-boost. Not substituted.

Before powered USB: unplug external battery AND disconnect/insulate OUT+ at J8.1.

Do not combine USB VBUS with converter output; main switch alone is insufficient.





TWO COPPER FACES / ALL HOLES UNPLATED / MANUAL LINKS ONLY / NOT RELEASED

Sheet: /
File: remote.kicad_sch

Title: Metal Monster / manual layer links

Size: A3	Date: 2026-10-14	**Rev: AR-8 DRAFT**
KiCad E.D.A. 9.0.2		Id: 1/1